ASSISTIVE AI & COMPUTER VISION PROJECT

APEC SMART GLASSES FOR THE BLIND

Real-Time Multimodal Vision AI, Local Vietnamese ASR & Tactile Accessibility

Repository: thanh1912-ut/APEC_mat_kinh...
Target SHA: d9ba1e0 (main)
Release: Android Pilot v0.4.0 (Code 8)
Core: Zipformer RNN-T + Gemini 3.8 Flash

< 180 ms	100% Tactile	Zero Audio Leak	651 Test Cases
LOCAL SPEECH STT LATENCY	HARDWARE VOLUME KEY UX	SPEECH STAYS ON SERVER	322 PYTEST + 329 JEST SUITES

1 MISSION, GOALS & TARGET SCOPE

APEC Smart Glasses is an assistive intelligence platform engineered specifically for blind and visually impaired users. It removes visual dependency through audio-spatial reasoning and tactile controls.

- **True Non-Visual UX:** Eliminates error-prone touchscreens; maps interactions directly to hardware volume buttons and acoustic earcons.
- **Privacy-Preserving Speech:** Streaming speech-to-text runs locally via Zipformer; raw human voice audio is never transmitted to cloud APIs.
- **Appliance Guidance (v0.4.0):** Interactive tactile assistance to locate, identify, and verify buttons on remotes, ACs, and appliances.
- **Spatial Vision Roadmap:** Pre-engineered architecture to interface Direct Time-of-Flight (dToF) LiDAR for 3D hazard avoidance.

2 SYSTEM ARCHITECTURE & PIPELINE

A Mobile Client: React Native / Expo captures rear camera JPEG (≤1280px) and voice query in M4A/AAC via volume keys.

B FastAPI Gateway: Validates SHA-256 Bearer Token, enforces 4 MiB limits, and queues into memory WorkGate (max 2 tasks).

C Local ASR Worker: PyAV resamples audio to mono 16 kHz; sherpa-onnx runs Vietnamese Zipformer RNN-T in memory.

D Vision Reasoning: JPEG image and text transcript queried to Gemini 3.8 Flash / 3.1 Flash Lite with `store=false`.

E Speech Delivery: Structured JSON outputs read out via Android TTS and earcons without TalkBack clash.

Component	Technology Stack	Key Specification
Mobile App	React Native / Expo 57, TS	Native Volume Key Module
Backend API	FastAPI, PyAV, Uvicorn	In-Memory Concurrency Gate
Local STT	sherpa-onnx (Zipformer)	51MB fp16 ONNX, ~60-180ms
Vision AI	Gemini 3.8 Flash / Lite	Structured JSON, store=false
Guidance DB	PostgreSQL + Redis	CAS Lock & Verification State

3 KEY TECHNICAL IMPLEMENTATIONS

- UX **Hardware Volume Key Controls:** Short-press *Volume Up* arms voice input (beeps and listens); second press captures image and sends request. Short-press *Volume Down* instantly stops speech readout or aborts pending requests. Long press preserves Android OS volume adjustments.
- PRIVACY **Local Streaming Speech Recognition:** Integrates `hynt/Zipformer-30M-RNNT-Streaming-6000h`. Speech packets are processed in temporary RAM buffers; audio is deleted immediately after inference without disk retention.
- AI AGENT **Verified Remote Appliance Guidance:** Enforces tactile anchors and visual verification gates before guiding next steps (e.g. power button vs mode switch), avoiding hazardous AI hallucinations.
- HARDWARE **ToF LiDAR Sensor Extension:** STMicroelectronics VL53L9CX Direct Time-of-Flight sensor with 54×44 matrix (2,376 distance zones) for millimeter-level obstacle proximity mapping.

4 ENGINEERING RIGOR & VISUAL ASSETS

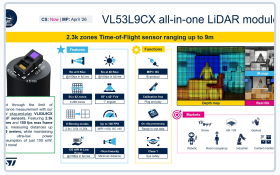
Verification Suite: 322 automated Python tests (WebRTC, tokens, WorkGate) + 329 Jest mobile tests across 17 suites with zero regressions.



Assistant UI



Settings UI



VL53L9CX 2376-Zone ToF

Security Architecture: Zero API keys in client APK; tokens stored in Expo SecureStore with SHA-256 verification. Strict single-turn concurrency gate prevents server DoS.